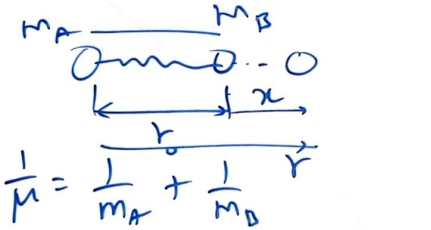
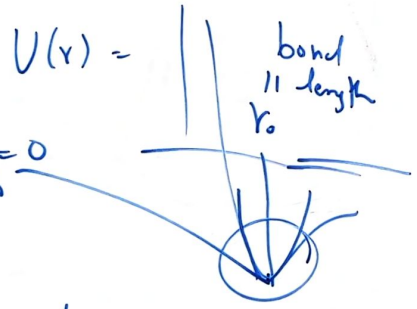
8-7-26

Molecular vibrations



$$\frac{1}{\mu} = \frac{1}{m_A} + \frac{1}{m_B}$$



$$\left.\frac{dU}{dr}\right|_{r=r_0} = 0$$

$$U(r) \simeq U(r_0) + \frac{1}{2}\underbrace{U''(r_0)}_{k}(r-r_0)^2$$

$$F = -\frac{dU}{dr} = -k(r-r_0)$$

$$\frac{d^2r}{dt^2} = \frac{d^2}{dt^2}(r_0 + x) = \frac{d^2x}{dt^2}$$

$$\mu\frac{d^2r}{dt^2} = \mu\frac{d^2x}{dt^2} = -k\underbrace{(r-r_0)}_{x}$$

$$\frac{d^2x}{dt^2} = -\left(\frac{k}{\mu}\right)x \qquad \omega^2$$

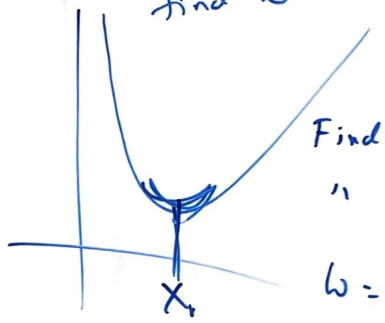
$$x = A\cos(\omega t + \alpha)$$

$$r = r_0 + A\cos(\omega t + \alpha)$$

Ex. Given A, B, m

$$U = \frac{A}{x} + Bx$$

find ω.



Find X_0

" $U''(X_0) = k$

$$\omega = \sqrt{\frac{k}{\mu}}$$

$$U(r) = \frac{A}{r^{12}} - \frac{B}{r^6}$$

LJ- potential

A, B, m

B x

ω.

Find X_0

" $U''(X_0) = k$

$\omega = \sqrt{\frac{k}{m}}$

$= \frac{B}{\gamma b}$

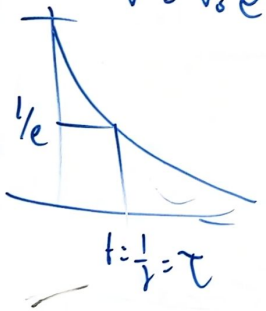
Damped motion

$$m\frac{dv}{dt} = -bv$$

$$\frac{dv}{dt} = -\gamma v, \quad \gamma = \frac{b}{m}$$

$$\int_{V_0}^{V} \frac{dv}{v} = -\gamma \int_0^t dt$$

$$V = V_0 e^{-\gamma t} = V_0 e^{-t/\tau}$$





$F = 6\pi\eta r v$

$\propto v$

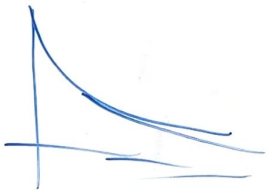
$= bv$

$$\frac{dx}{dt} = v = V_0 e^{-\gamma t}$$

$$\int_{x_0}^{x} dx = V_0 \int_0^t e^{-\gamma t}\, dt$$

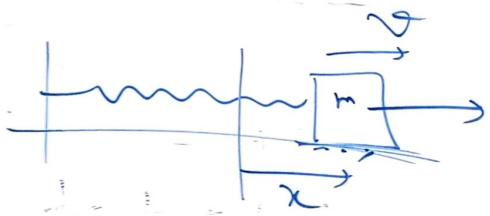
$$K = \frac{1}{2}mv^2 = \frac{1}{2}mV_0^2 e^{-2\gamma t}$$

$$= K_0 e^{-2t/\tau}$$



8-7-26

Damped oscillator



$$F_f = -bv = -b\frac{dx}{dt}$$

$$m\frac{d^2x}{dt^2} = \underbrace{-kx}_{F_{spring}} \underbrace{-bv}_{F_{damping}}$$

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$$

$\frac{k}{m} = \omega_0^2$

$\sqrt{\frac{k}{m}} = \omega_0$ = natural freq

$\frac{b}{m} = \gamma$ = damping const.

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

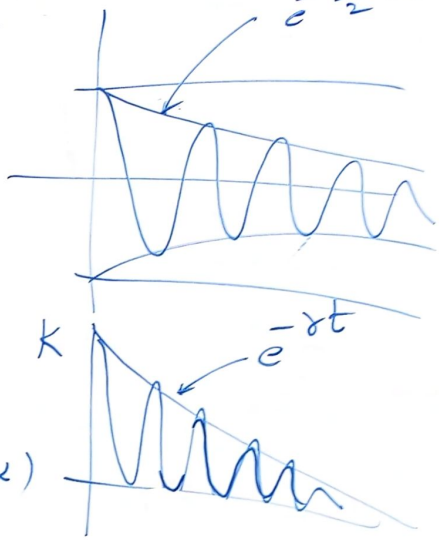
$x = Ce^{\lambda t} \longrightarrow$

AW $\quad x(t) = A\, e^{-\frac{\gamma}{2}t}\cos(\omega t + \varphi)$

$$\omega^2 = \omega_0^2 - \left(\frac{\gamma}{2}\right)^2$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2} = \omega_0\left[1 - \left(\frac{\gamma}{2\omega_0}\right)^2\right]^{1/2}$$

$$= \omega_0\left[1 - \left(\frac{1}{2\omega_0\tau}\right)^2\right]^{1/2}$$



$\langle K$

$\langle U$

$\langle E$

$P = \frac{d\langle E}{dt}$

$$\langle K \rangle \approx \frac{1}{4} m \omega_0^2 A^2 e^{-\gamma t}$$

$$\langle U \rangle \approx \frac{1}{4} m \omega_0^2 A^2 e^{-\gamma t}$$

$$\langle E \rangle \simeq \frac{1}{2} m \omega_0^2 A^2 e^{-\gamma t}$$

$$P = \frac{d\langle E \rangle}{dt} = -\gamma \langle E \rangle = \langle F v \rangle \quad \text{HW}$$

($F = -m\gamma v$)



Quiz 1 - 8.7.26



$m = 2, \quad k = 8$

$x(0) = 1, \quad v(0) = 2\sqrt{3}$

Find ω, A, α in

$x(t) = A \cos(\omega t + \alpha)$

Find $v(t)$, E

Find $x(2)$, $v(2)$,